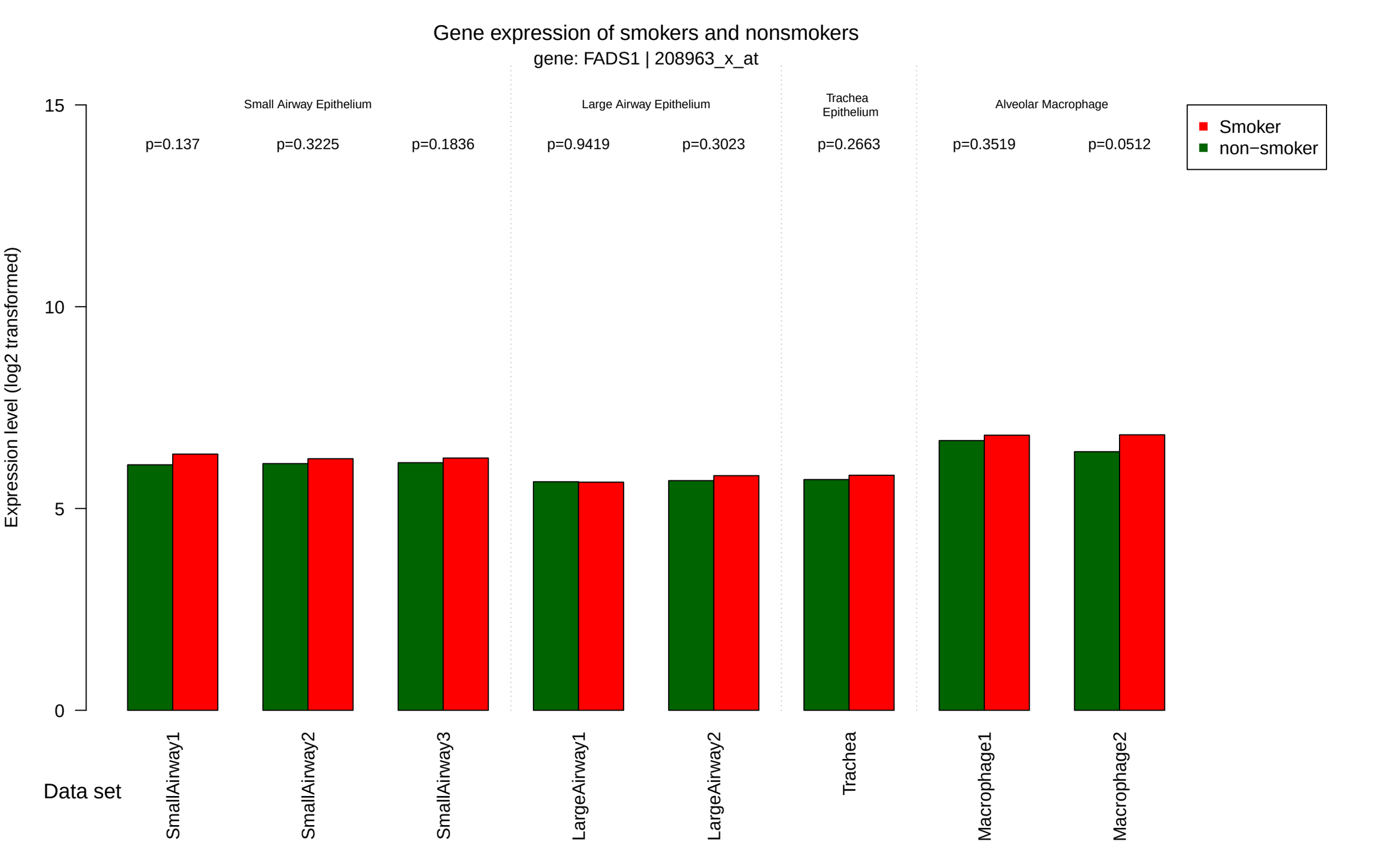
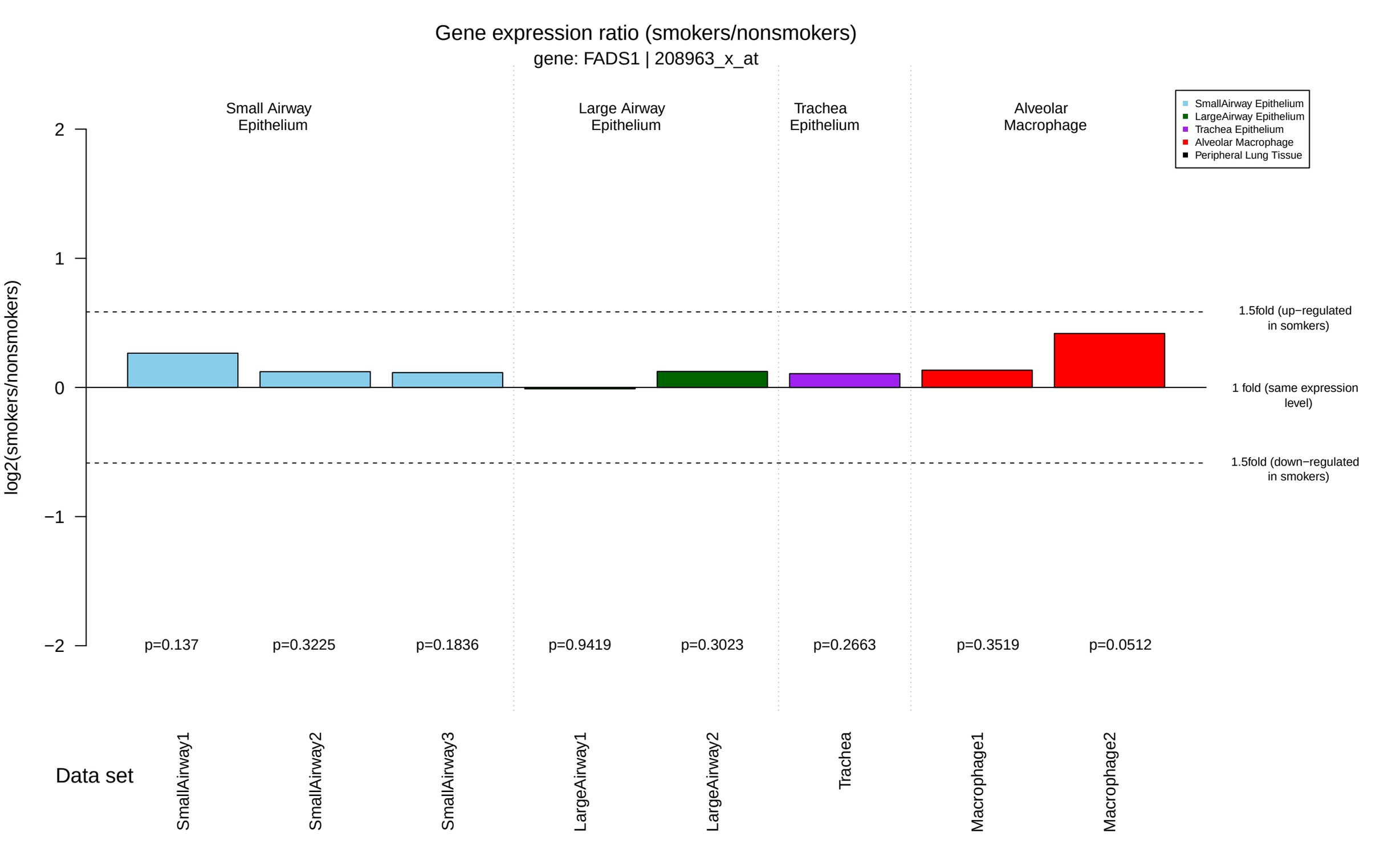
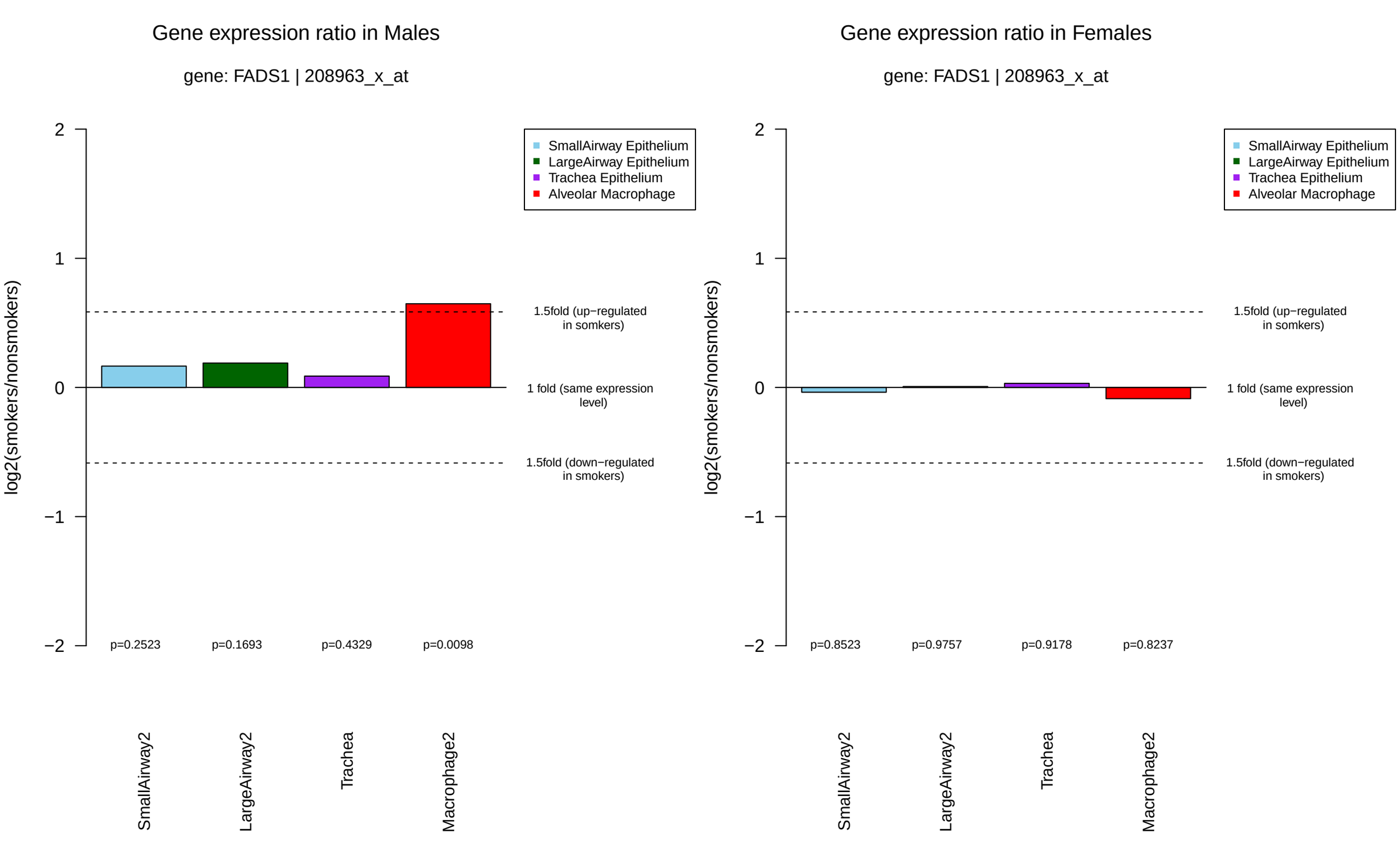
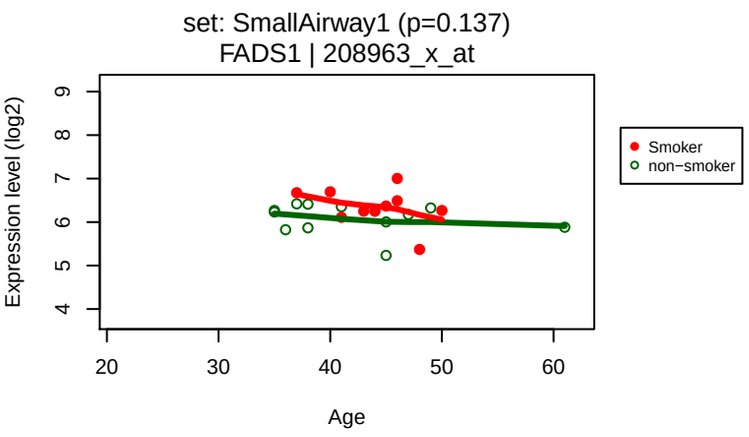
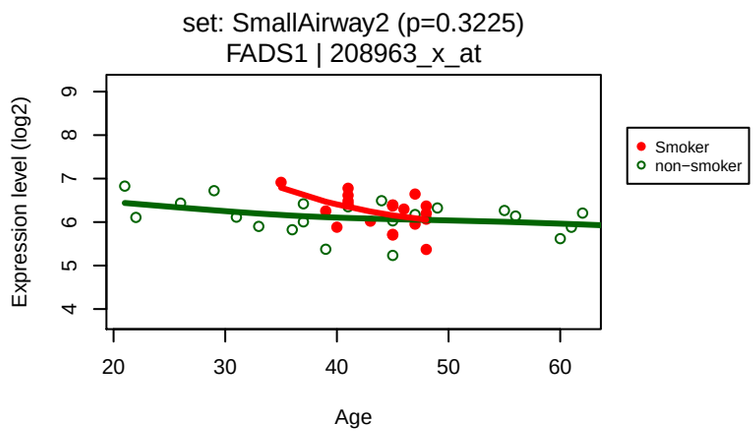
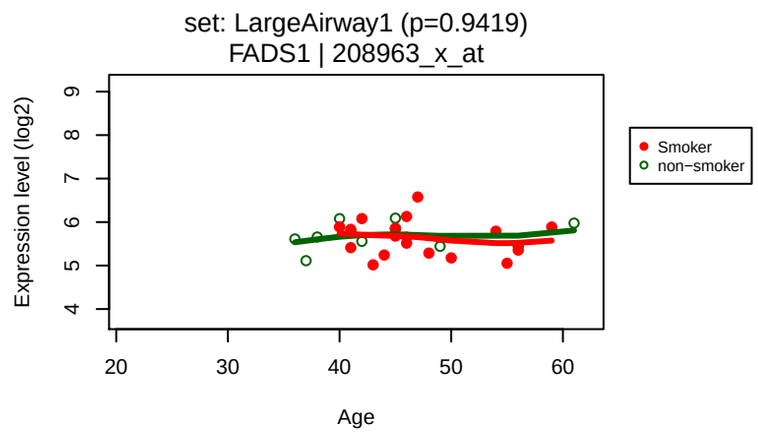

Gene expression of smokers and nonsmokers
gene: FADS1 | 208963_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.137
p=0.3225
p=0.1836
p=0.9419
p=0.3023
p=0.2663
p=0.3519
p=0.0512
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

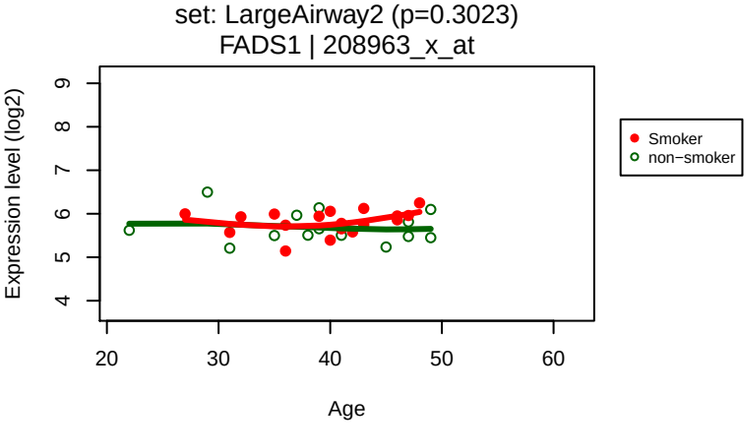

Gene expression ratio (smokers/nonsmokers)
gene: FADS1 | 208963_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.137
p=0.3225
p=0.1836
p=0.9419
p=0.3023
p=0.2663
p=0.3519
p=0.0512
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

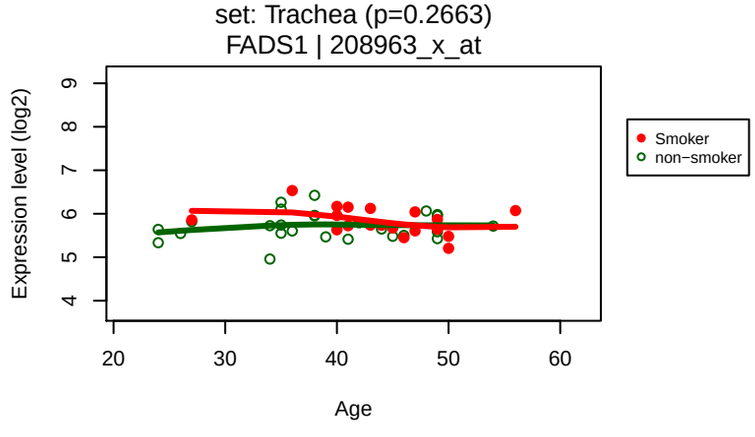

Gene expression ratio in Males
gene: FADS1 | 208963_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2523
p=0.1693
p=0.4329
p=0.0098
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FADS1 | 208963_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8523
p=0.9757
p=0.9178
p=0.8237
SmallAirway2
LargeAirway2
Trachea
Macrophage2

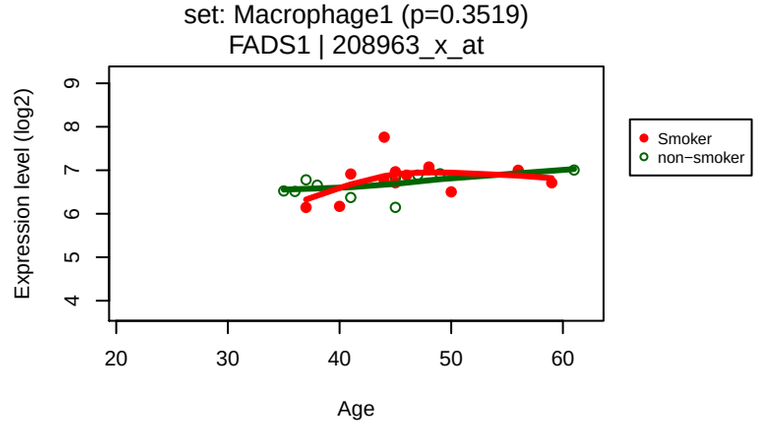

set: SmallAirway1 (p=0.137)
FADS1 | 208963_x_at
Expression level (log2)
Age
Smoker
non−smoker

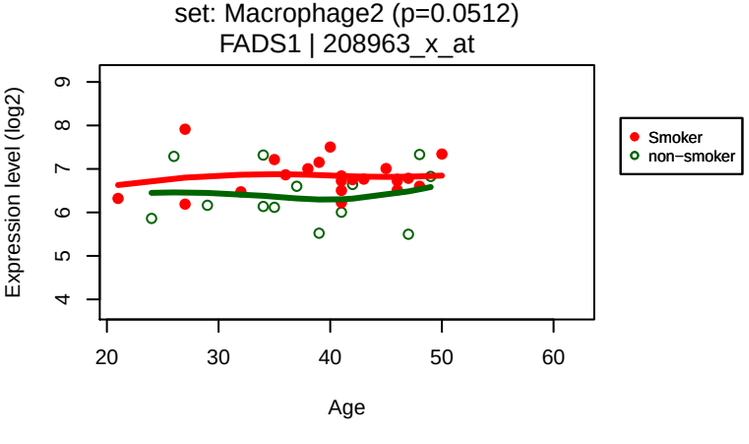

set: SmallAirway2 (p=0.3225)
FADS1 | 208963_x_at
Expression level (log2)
Age
Smoker
non−smoker

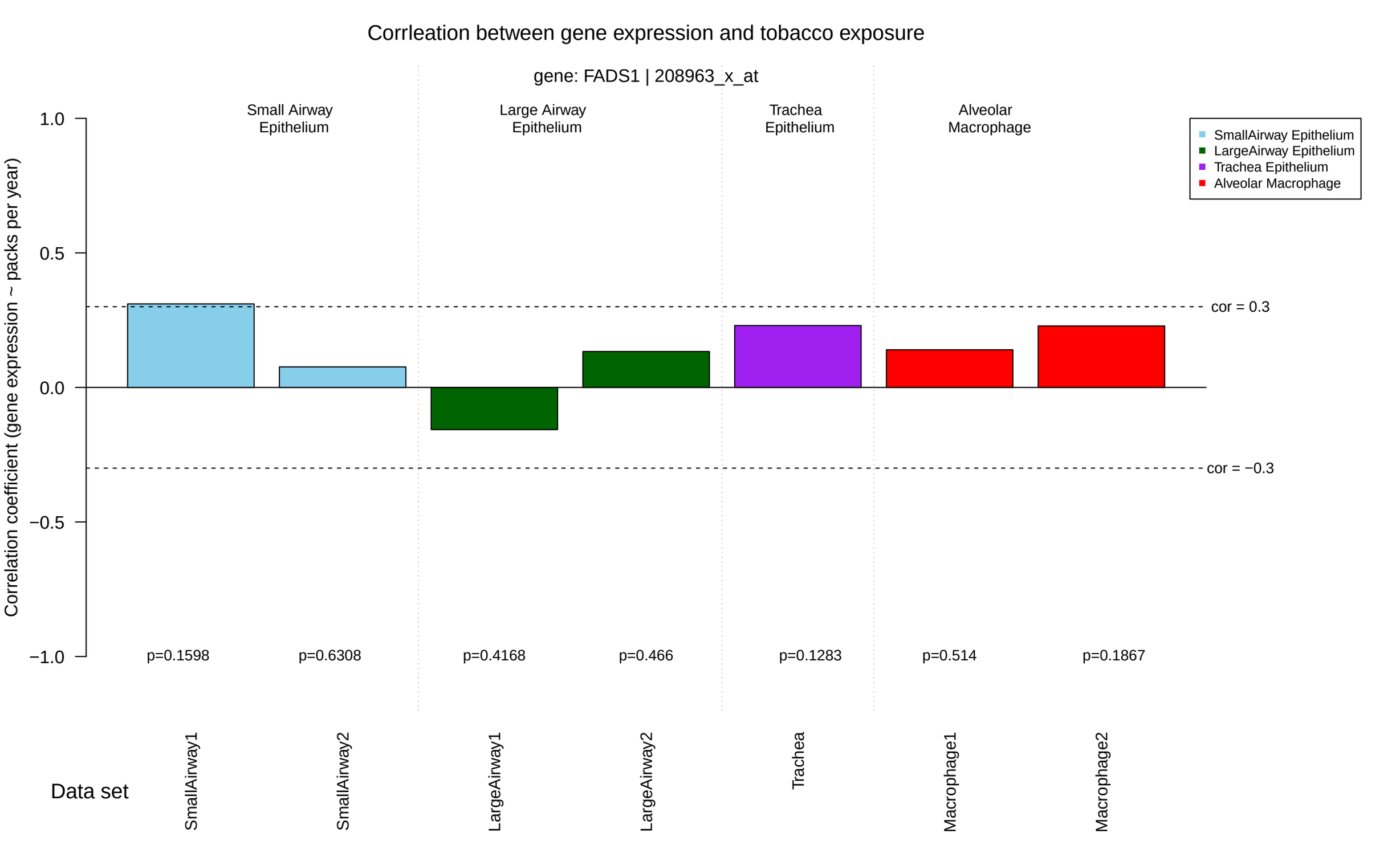

set: LargeAirway1 (p=0.9419)
FADS1 | 208963_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3023)
FADS1 | 208963_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2663)
FADS1 | 208963_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3519)
FADS1 | 208963_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0512)
FADS1 | 208963_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FADS1 | 208963_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1598
p=0.6308
p=0.4168
p=0.466
p=0.1283
p=0.514
p=0.1867
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2